

# Reproducible Experiments

Project: Responsible Plant Classifier

## 1. Your mission

### Your learning target

I can record enough detail to reproduce a run.

I can distinguish reproducibility from validity.

I can report all planned seeds instead of selecting the best.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

A result is reproducible only when another builder can recreate the data, split, code, settings, seed, and metric calculation.



LEARNING TARGET Two teams use the same random seed but different label files. Their results differ. List the records needed to reproduce the experiment.

Builder name and experiment date:

### Readiness for the connected explanation

What must accompany a reported validation score?

### Readiness for Versioned experiments

Why should expected test results be written before running the function?

#### My readiness explanation

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### This week's end goal

By the end, I can explain reproducible experiments using a traced example and a stated limitation.

## 2. Notice the evidence

PICTURE ENTRY

## Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



**MYSTERY** The team must use reproducible experiments to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

### A closer explanation

A seed can help repeat a random split or initialization, but it is not a magic quality stamp. Save data version, split membership, code version, settings, preprocessing, seed and metric definition. Some software operations remain nondeterministic, so document tolerances and environment too.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines experimental records and seeds.

### Apply the connected idea: Versioned experiments

A fictional activity scoring experiment compares two weight settings on the same development records. Save code version, input version, weights, tie rule and outputs. A label such as v2 is useful only when it

refers to an identifiable saved state.

### 3. See how it works

ML MECHANISM

**Input** → **Model** → **Evidence**

Trace the information, fitted behavior, and evaluation proof separately.

**INPUT**

Reproduce a peer's run from an experiment card.

**MODEL**

A result is reproducible only when another builder can recreate the data, split, code, settings, seed, and metric calculation.



**OUT**

## EVIDENCE

Reproduce a peer's run from an experiment card.

QUESTION Reproduce a peer's run from an experiment card.

MODEL IDEA A result is reproducible only when another builder can recreate the data, split, code, settings, seed, and metric calculation.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Record data=v2, split=grouped-s1, code=commit A7, seed=42. Record preprocessing, features, model type, settings, and library versions.

Draw a fresh data → model → prediction → evaluation trace:

### A closer explanation

A team declares seeds 11, 22, 33 before running. Validation accuracies are 14/20, 16/20, 12/20:70, 80, 60 percent. Mean accuracy is 70 percent, with a 20-percentage-point range. Reporting only seed 22 as the model result hides the declared experiment.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

### Apply the connected idea: Versioned experiments

```
items = [{"name": "A", "fun": 4, "learning": 1},
          {"name": "B", "fun": 2, "learning": 4}]

for version, fun_weight, learning_weight in [("v1", 2, 1), ("v2", 1, 2)]:
    scores = []
```

```
for item in items:

    scores.append(fun_weight * item["fun"]

                  + learning_weight * item["learning"])

print(version, scores)
```

## Follow the evidence

### Step or case · Evidence · Meaning

**Step or case:** v1 weights 2 and 1

**Evidence:** A 9; B 8

**Meaning:** A ranks first

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**Step or case:** v2 weights 1 and 2

**Evidence:** A 6; B 10

**Meaning:** B ranks first

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Read each row from left to right and connect the evidence to the stated rule.

The ranking changes because the preference weights changed. That is a policy comparison, not proof that v2 is more accurate. Keep the same data and constraints when comparing weights, and record why a policy better fits the stated goal.

## Compare one planned change at a time

For the worked records, v1 gives A a score of 2 times 4 plus 1, or 9, and B a score of 2 times 2 plus 4, or 8. v2 gives A 4 plus 2 times 1, or 6, and B 2 plus 2 times 4, or 10. The winner changes because the weights changed. Keep the records fixed so this explanation can be checked. A different preference policy is not automatically a more accurate model.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

# 4. Compare the cases

## EVALUATION DESIGN

### Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



## **NORMAL CASE**

Expected data from the intended setting



## **BOUNDARY CASE**

Value near a threshold or class boundary



## MISSING CASE

Unknown field, label, or necessary evidence



## SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

**RELEASE RULE** Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

## Change one thing in the pictured case

Change: The report deletes the 60% run.

Prediction: The remaining mean rises to 75%, but the reported comparison becomes selectively incomplete.

Reason: Removing an unfavourable declared run changes the evidence rather than improving the method.

## 5. Words that unlock the idea

### VOCABULARY

#### Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

## **DATA VERSION**

identified dataset state

3

## **RANDOM SEED**

value controlling repeatable  
randomness

4

## EXPERIMENT RECORD

complete inputs, settings,  
and outcomes

**Term · Precise meaning · My ML example**

**Term:** reproducibility

**Precise meaning:** ability to recreate a result

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

**Term:** data version

**Precise meaning:** identified dataset state

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

**Term:** random seed

**Precise meaning:** value controlling repeatable randomness

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

**Term:** experiment record

**Precise meaning:** complete inputs, settings, and outcomes

**My ML example:** \_\_\_\_\_

\_\_\_\_\_

MISCONCEPTION TO REPAIR Writing down the final accuracy makes an experiment reproducible.

Repair the claim using a definition, calculation, or test record:

### Words used in the closer explanation

#### Word · Meaning

**Word:** Seed

**Meaning:** An initial value controlling a pseudo-random process

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**Word:** Run record

**Meaning:** A saved account of inputs, settings and outputs

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**Word:** Reproducibility

**Meaning:** Ability to repeat a specified procedure

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**Word:** Cherry-picking

**Meaning:** Reporting favorable results while hiding relevant others

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### Terms for Versioned experiments

#### Term · Meaning

**Term:** Version

**Meaning:** An identified state of code or rules

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**Term:** Configuration

**Meaning:** Settings used for a run

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**Term:** Development set

**Meaning:** Cases used while choosing or improving a method

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**Term:** Holdout

**Meaning:** Cases reserved for an evaluation after choices are fixed

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## 6. Follow a worked example

WORKED EVIDENCE

### Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

#### WEEK 26 • WORKED EVIDENCE

- 1 Record data=v2, split=grouped-s1, code=commit A7, seed=42.
- 2 Record preprocessing, features, model type, settings, and library versions.
- 3 Peer reruns and gets 16/20 correct, matching 80% accuracy.
- 4 A mismatch triggers environment and data comparison—not silent replacement.

STEP 1 Record data=v2, split=grouped-s1, code=commit A7, seed=42.

STEP 2 Record preprocessing, features, model type, settings, and library versions.

STEP 3 Peer reruns and gets 16/20 correct, matching 80% accuracy.

STEP 4 A mismatch triggers environment and data comparison—not silent replacement.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

### A closer explanation

### Experimental records and seeds at a glance

## Part · Evidence or action · Interpretation

**Part:** Before running

**Evidence or action:** Declare 11 22 33

**Interpretation:** Save data and settings

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**Part:** Observed results

**Evidence or action:** 70 80 60 percent

**Interpretation:** Keep all three runs

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**Part:** Summary

**Evidence or action:** Mean 70 percent

**Interpretation:** Range 20 points

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The seed is part of the procedure, not evidence that a split represents the target population. Reproducing leakage perfectly reproduces a mistake. Check group separation and legitimate features before celebrating identical outputs.

**A file contains only seed 22 and 80 percent. Name four missing items. Explain the steps or evidence behind your response.**

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## Apply the connected idea: Versioned experiments

**If the data also changes between v1 and v2, can the ranking change be attributed only to weights?**

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## Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Reproducible Experiments

Record variation across declared seeds instead of selecting the prettiest run.

#### The facts we will use

Same fixed validation set of 10 cases; declared runs seed 4=6 correct, seed 5=8, seed 6=7. All three are reported.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/26

### All declared seeds

accuracy percent



Bar length shows the supplied value; the marked baseline is zero.

# Reproducible Experiments

## Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



### 1. Keep run identities

Record seeds 4, 5, 6 with 6/10, 8/10, 7/10.



### 2. Summarise variation

Mean accuracy is 70%; range is 60% to 80%.



### 3. Preserve context

Record data version, split, code, settings and seeds, not just the winning 80%.

## CHANGE ONE THING

The report deletes the 60% run.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Keep run identities

**What happens and why:** Record seeds 4, 5, 6 with 6/10, 8/10, 7/10.

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**Step:** 2. Summarise variation

**What happens and why:** Mean accuracy is 70%; range is 60% to 80%.

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### Step: 3. Preserve context

**What happens and why:** Record data version, split, code, settings and seeds, not just the winning 80%.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Read the picture in order. Record data version, split, code, settings and seeds, not just the winning 80%. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

Seeds help reproduction but do not necessarily remove hardware or library nondeterminism; report the actual environment.

Read the labelled picture: What should accompany the mean 70%? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: The report deletes the 60% run. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

ERROR DETECTIVE

### Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



#### COMPARE • DIAGNOSE • REPAIR

Tempting claim: Writing down the final accuracy makes an experiment reproducible. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** passes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** defined behavior

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** missing case

**Expected:** safe fallback

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** shifted case

**Expected:** measured limit

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**CLAIM TO CHALLENGE** Writing down the final accuracy makes an experiment reproducible.

Diagnose one cause and name the cleanest next experiment:

## A closer explanation

When comparing two methods, use the same declared splits or seeds where appropriate and compare paired results. A mean across reused samples does not automatically provide a statistical confidence claim. Report the procedure and observed variation plainly.

**One planned seed fails before prediction. How should it appear in the record?**

\_\_\_\_\_  
\_\_\_\_\_

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## Apply the connected idea: Versioned experiments

Once holdout results guide another change, those cases have become development evidence. A later independent evaluation needs new reserved cases. Small holdouts still provide limited evidence.

**You revise the policy after reading final-case failures. Are those cases still untouched holdout evidence?**

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## 8. Practise together

ML STUDIO

### Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



#### **FREEZE THE EXPERIMENT**

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

### Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

## A closer explanation

A small record can contain run ID, timestamp, dataset checksum, training and validation IDs, model settings, feature version, seed, predictions and notes. Keep personal information out of exported logs. Save failed runs with reasons instead of quietly removing them.

## Apply the connected idea: Versioned experiments

Overwriting a file called latest removes the ability to reproduce earlier results unless versions are retained elsewhere. Save identifiable versions and a run log. Avoid editing a reserved final test after seeing its output to make the chosen version look better.

## Practise with less help: Improve a record

Use workbook W2 for the connected example “Experimental records and seeds”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Check your reasoning

Use the worked case for Experimental records and seeds. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

## Practise the connected topic: Versioned experiments

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

What should accompany the mean 70%? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Reproduction Swap: exchange incomplete experiment cards, identify missing fields, then reproduce and explain any mismatch.

- 1. Write the hypothesis and expected evidence before running.
- 2. Reproduction Swap: exchange incomplete experiment cards, identify missing fields, then reproduce and explain any mismatch.
- 3. Run one normal, one boundary, one missing, and one shifted case.
- 4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

**Case · Prediction · Actual · Calculation / evidence · Meaning / next test**

**Case:** Normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Case:** Shifted

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Calculation / evidence:** \_\_\_\_\_

**Meaning / next test:** \_\_\_\_\_

\_\_\_\_\_

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

### MASTERY

#### From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



**ML BUILDER MASTERY**  
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

**DEFINE** Use all four terms precisely: reproducibility, data version, random seed, experiment record.

**CALCULATE** Reproduce the worked evidence with units and denominator.

**TRACE** Explain the data, code or decision path.

**DIAGNOSE** Writing down the final accuracy makes an experiment reproducible.

**TRANSFER** Solve a fresh dataset or model case.

**IMPROVE** Change one cause and rerun fixed evidence.

**DEFEND** State intended use, metric, limitation, and human responsibility.

**MASTERY RULE** A coached calculation becomes mastery only after a different case is completed independently.

## 11. Connect your project

### CUMULATIVE PROJECT

#### Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.



## RESPONSIBLE PLANT CLASSIFIER

Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, predictions, and metrics.

**THIS WEEK'S CHECKPOINT** Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, predictions, and metrics.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

**RESPONSIBLE RELEASE** Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

### Evidence for your project

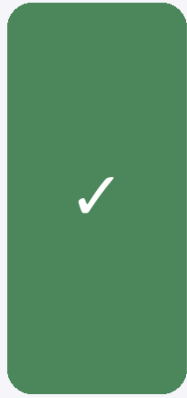
For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

## 12. Show what you know

INDEPENDENT PROOF

**Explain • Calculate • Transfer • Defend**

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



### INDEPENDENT ML BUILDER PROOF

Reproduce a peer's run from an experiment card.

REQUIRED PROOF Reproduce a peer's run from an experiment card.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Overfitting.

## 13. Study a complete case

### Week 26 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

A report says accuracy85% without data split, code version, seed or preprocessing.

#### The question

Specify reproducibility records.

#### Worked explanation

Record data identity, group/split IDs, code/dependency versions, preprocessing, features, settings, seeds where relevant, metric definition and raw predictions. Seed alone is insufficient.

**Explain it in your own words**

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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**Trace the supplied case visually**

**Make an experiment reproducible**

| Evidence or stage · Value or result |
|-------------------------------------|
|-------------------------------------|

**Evidence or stage:** Data

**Value or result:** Identity and split/group IDs

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**Evidence or stage:** Code

**Value or result:** Version and dependencies

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**Evidence or stage:** Procedure

**Value or result:** Features, preprocessing and settings

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**Evidence or stage:** Evidence

**Value or result:** Seeds, metric definitions and raw predictions

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A report says accuracy85% without data split, code version, seed or preprocessing.

Specify reproducibility records.

Record data identity, group/split IDs, code/dependency versions, preprocessing, features, settings, seeds where relevant, metric definition and raw predictions. Seed alone is insufficient.

# 14. Try a new case independently

## Independent case: Apply independently

Declared seeds 1, 2, 3 produce 9/12, 8/12, 10/12 correct. Compute the mean accuracy and range.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## Week 26 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Two runs use different random splits and different feature sets. Their scores differ. What must be recorded, and why is the cause unclear?

My answer, with a trace, calculation or supporting evidence

### Write or draw your reasoning and evidence.

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## Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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## Connect to your project

Give every Plant Classifier experiment an ID and immutable record linking dataset, split, code, settings, predictions, and metrics.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

## Transfer practice from the original programme

Two teams use the same random seed but different label files. Their results differ. List the records needed to reproduce the experiment.

**Explain the mechanism, evidence and one limit.**

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**Use feedback to improve**

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

**15. Your lesson route**

Start with this question: Record variation across declared seeds instead of selecting the prettiest run.

**By the end, I can · How I will show it**

**By the end, I can:** Record enough detail to reproduce a run

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Distinguish reproducibility from validity

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Report all planned seeds instead of selecting the best

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

**16. Finish and return**

Close your notes. Explain this goal in your own words: Record enough detail to reproduce a run

**My deciding step, evidence and one remaining question**

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.